



Computing with Diffractions

An LCD-Based Optical Neural Network for Handwritten Digit Classification

Student: Chan Yu Sing · Supervisor: Prof. Kenneth K.Y. Wong

“What if laser and LCD screen could replace a GPU?”

The Problem

Training and running large neural networks is increasingly energy-intensive. Data centres are projected to consume 1,000 TWh of electricity per year by 2026 [1] — comparable to the entire annual output of several power plants. The core bottleneck is not just energy: conventional computers process data sequentially, and memory bandwidth constrains how fast information can flow between storage and computation. Every neural network inference requires billions of transistor switches to fire, all consuming power and generating heat.

The Physics

Light offers a fundamentally different computing substrate. A coherent beam travels at 3×10^8 m/s and diffracts through apertures according to well-understood wave physics. When a 200×200 LCD modulates the amplitude of a beam, it performs 40,000 independent multiplications simultaneously — in a single exposure, passively, with no electronic switching at all. The computation happens as the wavefront propagates through space.

Why LCD?

This work uses an off-the-shelf LCD panel — cheap enough for a student lab, fast enough to reprogram in seconds.

Technology	Cost / layer
SLM	> \$10 000
Transparent Amplitude LCD	~ \$10

System Architecture

The key insight is that no computation happens electronically during the optical forward pass. Once the laser illuminates the first LCD, the physics of wave diffraction carries the information through space. The second LCD mask applies a second learned transformation. By the time light reaches the detector, the digit classification is encoded in the intensity pattern — readable with a photodetector array.

Hardware Specifications

Parameter	Value
Laser wavelength	532 nm
LCD resolution	200×200 pixels
Pixel pitch 138.3 μm	138.3 μm
Inter-layer spacing	50 mm

Watching the Diffraction Happen

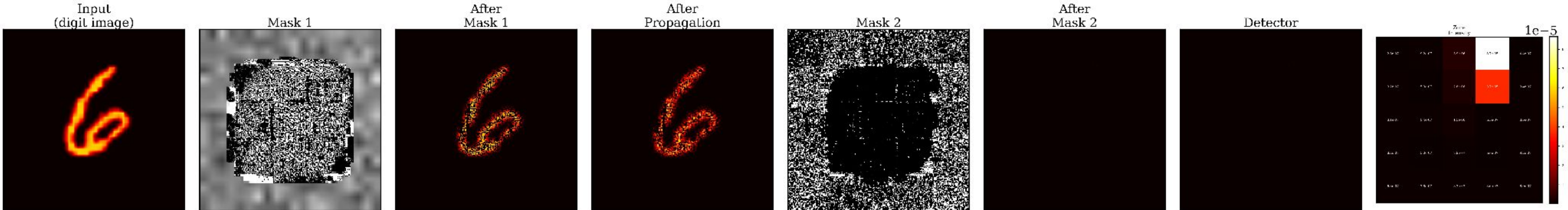


Fig. 2. Light propagation through the two-layer ONN for digit 6.

Detection Algorithm

After propagating through both masks, the light arrives at a flat detector plane as a 2D intensity image. The detector is divided into a 5×5 grid of 25 zones, each assigned to one of the ten digit classes (with some classes sharing zones). The average light intensity within each zone is measured, and the digit class corresponding to the zone with the highest average intensity is taken as the network’s prediction. This max-zone readout requires no digital computation — it can in principle be implemented with a simple photodetector array.

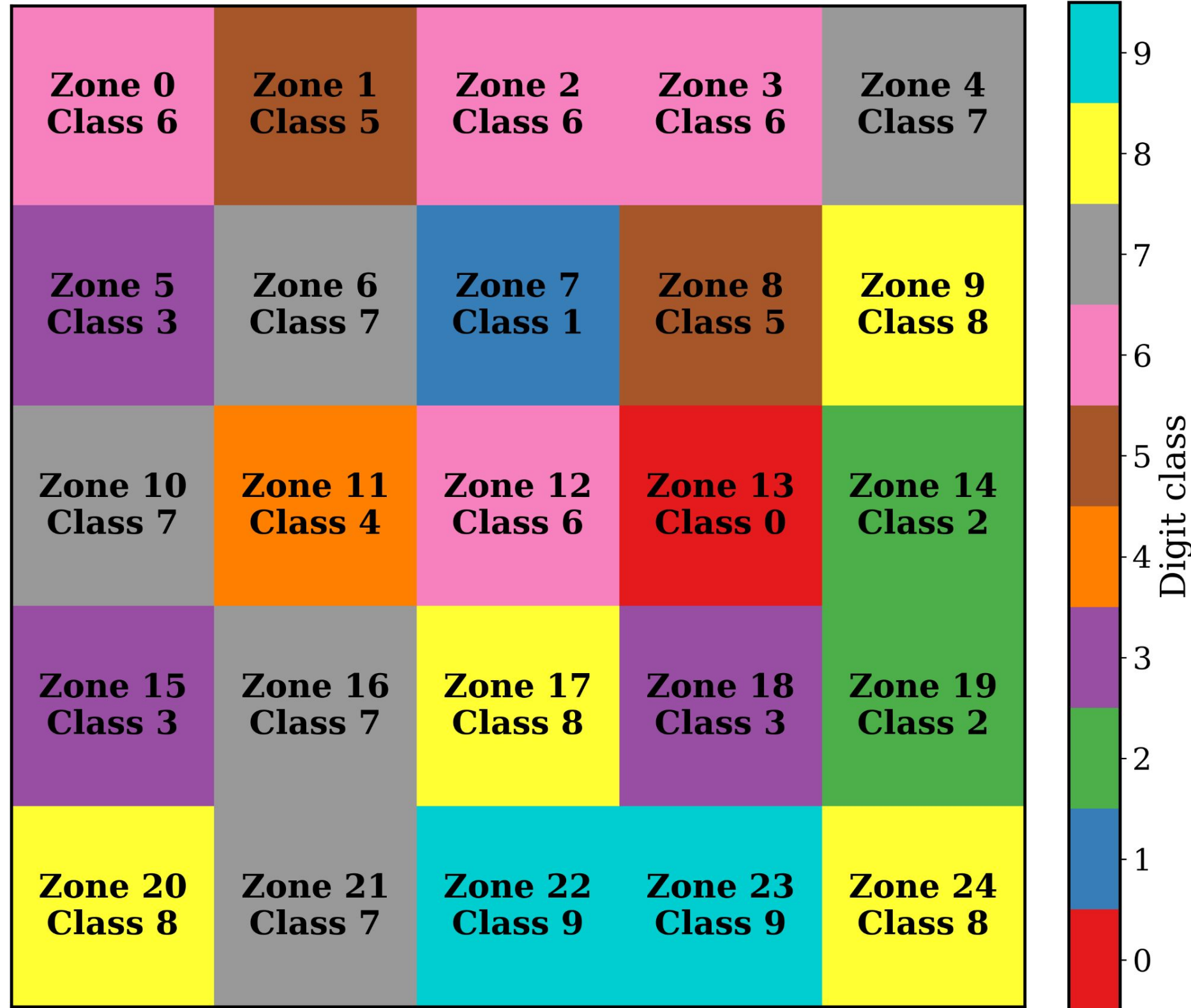


Fig. 3. Max-zone 5×5 detector zone-to-class assignment map.

Classification Performance

90.85% test accuracy on 10,000 MNIST images
Independently verified in **MATLAB**

Matches the 5-layer phase benchmark [2] at 91.75% — using 2 layers and hardware costing 200× cheaper. The result was independently reproduced in MATLAB from scratch to rule out implementation artefacts. The reproduction agree with the PyTorch training result.

Note: results obtained in simulation; physical hardware assembly is the next step.

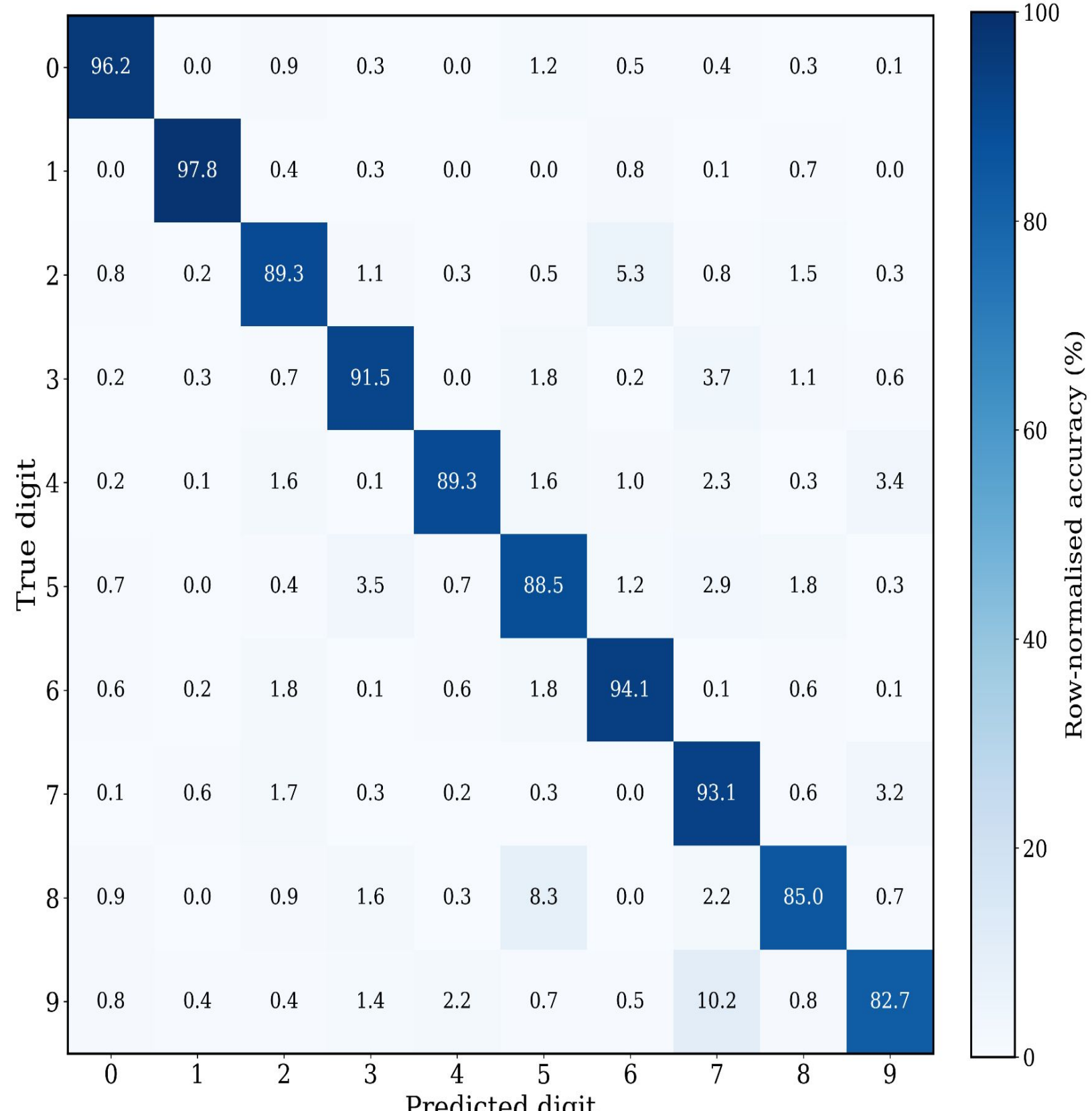


Fig. 5. Row-normalised confusion matrix for the max-zone 5×5 model on the 10,000-image MNIST test set.

System Layers	Modulation	Accuracy
Lin et al. [2]	5 layers Phase	91.75%
This work	2 layers Amplitude	90.85%

Conclusions

Optical computing has attracted renewed interest as an energy-efficient pathway to AI inference. This work demonstrates that a two-layer amplitude-modulation Optical Neural Network built from consumer LCD panels can match the performance of significantly more complex and expensive phase-based systems. A complete two-layer ONN system — laser, two LCD panels, detector — can be assembled for under USD 100 and retrained for a new task in without replacing any hardware.

Future: Physical hardware assembly

References

[1] International Energy Agency, "Electricity 2024: Analysis and Forecast to 2026," IEA, Jan. 2024.
[2] X. Lin et al., "All-optical machine learning using diffractive deep neural networks," Science, vol. 361, pp. 1004–1008, 2018.